

● MEDIUM

● INFORMATION AND IDEAS

---

# Information and Ideas

09

10 questions · ~15 min

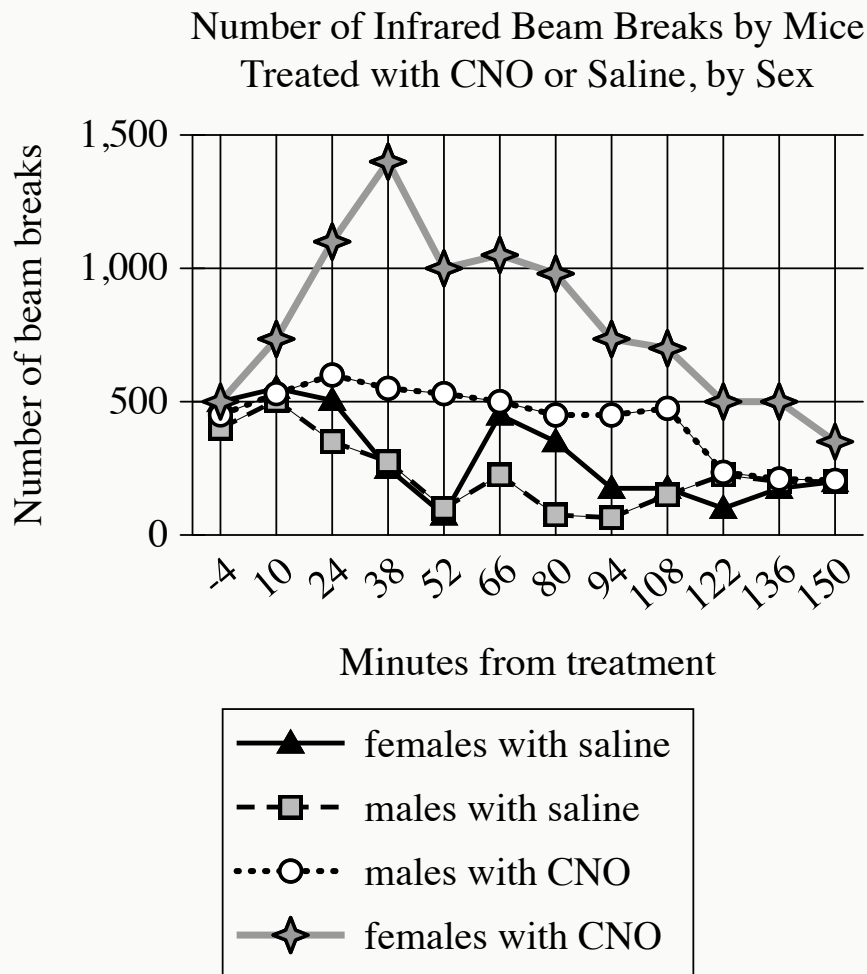
Q1

INFORMATION AND IDEAS

Astronomers investigated the Arabia Terra region of Mars because it appears to contain irregularly shaped craters that may have been caused by massive volcanic explosions. In their investigations of Arabia Terra, the researchers found remnants of ash deposits in an amount and thickness that would result from a massive volcanic eruption. However, erosion and past resurfacing events could have modified the surface of the planet. Therefore, \_\_\_\_\_blank

Which choice most logically completes the text?

- A. the current makeup of the Arabia Terra region might not accurately reflect the volcanic activity of Mars's past.
- B. eruptions from Mars's volcanoes were likely not as massive as astronomers previously believed.
- C. ash was most likely expelled from multiple different volcanoes on Mars's surface.
- D. the craters found in the Arabia Terra region were necessarily created by events other than volcanic eruptions.



- All values are approximate.
- The following 4 lines are shown:
  - females with CNO
  - females with saline
  - males with CNO
  - males with saline
- The females with CNO line:
  - Begins at negative 4, 500
  - Rises sharply to 10, 735
  - Rises sharply to 24, 1,100
  - Rises sharply to 38, 1,400

- Falls sharply to 52, 1,000
- Rises gradually to 66, 1,050
- Falls gradually to 80, 980
- Falls sharply to 94, 735
- Falls gradually to 108, 700
- Falls sharply to 122, 500
- Remains level to 136, 500
- Falls sharply to 150, 350
- The females with saline line:
  - Begins at negative 4, 500
  - Rises gradually to 10, 550
  - Falls gradually to 24, 505
  - Falls sharply to 38, 250
  - Falls sharply to 52, 75
  - Rises sharply to 66, 450
  - Falls gradually to 80, 350
  - Falls sharply to 94, 175
  - Remains level to 108, 175
  - Falls gradually to 122, 100
  - Rises gradually to 136, 175
  - Rises gradually to 150, 200
- The males with CNO line:
  - Begins at negative 4, 450
  - Rises gradually to 10, 530
  - Rises gradually to 24, 600
  - Falls gradually to 38, 550
  - Falls gradually to 52, 530
  - Falls gradually to 66, 500
  - Falls gradually to 80, 450
  - Remains level to 94, 450
  - Rises gradually to 108, 475
  - Falls sharply to 122, 235
  - Falls gradually to 136, 210
  - Falls gradually to 150, 205
- The males with saline line:
  - Begins at negative 4, 400
  - Rises sharply to 10, 505
  - Falls sharply to 24, 350

- Falls gradually to 38, 275
- Falls sharply to 52, 100
- Rises sharply to 66, 225
- Falls sharply to 80, 75

To investigate the influence of certain estrogen-responsive neurons on energy expenditure, biologist Stephanie Correa et al. treated female and male mice with either saline solution or clozapine-N4-oxide (CNO), which activates the neurons. Monitoring the activity levels of the mice by measuring how frequently the animals broke infrared beams crossing their enclosures, Correa et al. found that the mice in their study showed sex-specific differences in response to neuron activation:

\_\_\_\_\_blank

Which choice most effectively uses data from the graph to complete the assertion?

- A. the four groups of mice differed greatly in their activity levels before treatment but showed identical activity levels at the end of the monitoring period.
- B. saline-treated females showed substantially more activity at certain points in the monitoring period than saline-treated males did.
- C. CNO-treated females showed more activity relative to saline-treated females than CNO-treated males showed relative to saline-treated males.
- D. CNO-treated females showed a substantial increase and then decline in activity over the monitoring period, whereas CNO-treated males showed a substantial decline in activity followed by a steep increase.

To understand how Paleolithic artists navigated dark caves, archaeologist M<sup>a</sup> Ángeles Medina-Alcaide and her team tested different lighting methods in a cave in Spain using replicas of artifacts found in European caves with art. They used three different Paleolithic light sources—torches, animal-fat lamps, and fireplaces—determining that each likely had a specific purpose. For instance, the team learned that the animal-fat lamps were less useful than torches while walking because the lamps didn't illuminate the cave floor.

Which choice best states the main idea of the text?

- A. Medina-Alcaide and her team's study demonstrated that fireplaces were essential to the creators of Paleolithic cave art.
- B. Medina-Alcaide and her team discovered that Paleolithic cave artists in Spain used animal-fat lamps more often than they used torches.
- C. Medina-Alcaide and her team were reluctant to draw many conclusions from their study because of the difficulty they had replicating light sources based on known artifacts.
- D. Medina-Alcaide and her team tested Paleolithic light sources and learned some details about how Paleolithic artists traveled within dark caves.

In her 2021 article “Throwaway History: Towards a Historiography of Ephemera,” scholar Anne Garner discusses John Johnson (1882–1956), a devoted collector of items intended to be discarded, including bus tickets and campaign pamphlets. Johnson recognized that scholarly institutions considered his expansive collection of ephemera to be worthless—indeed, it wasn’t until 1968, after Johnson’s death, that Oxford University’s Bodleian Library acquired the collection, having grasped the items’ potential value to historians and other researchers. Hence, the example of Johnson serves to \_\_\_\_\_ blank

Which choice most logically completes the text?

- A. demonstrate the difficulties faced by contemporary historians in conducting research at the Bodleian Library without access to ephemera.
- B. represent the challenge of incorporating examples of ephemera into the collections of libraries and other scholarly institutions.
- C. lend support to arguments by historians and other researchers who continue to assert that ephemera holds no value for scholars.
- D. illustrate both the relatively low scholarly regard in which ephemera was once held and the later recognition of ephemera’s possible utility.

Approximate Rates of Speech and Information Conveyed for Five Languages

| Language   | Rate of speech<br>(syllables per second) | Rate of information conveyed (bits per second) |
|------------|------------------------------------------|------------------------------------------------|
| Serbian    | 7.2                                      | 39.1                                           |
| Spanish    | 7.7                                      | 42.0                                           |
| Vietnamese | 5.3                                      | 42.5                                           |
| Thai       | 4.7                                      | 33.8                                           |
| Hungarian  | 5.9                                      | 34.6                                           |

A group of researchers working in Europe, Asia, and Oceania conducted a study to determine how quickly different Eurasian languages are typically spoken (in syllables per second) and how much information they can effectively convey (in bits per second). They found that, although languages vary widely in the speed at which they are spoken, the amount of information languages can effectively convey tends to vary much less. Thus, they claim that two languages with very different spoken rates can nonetheless convey the same amount of information in a given amount of time.

Which choice best describes data from the table that support the researchers' claim?

- A. Among the five languages in the table, Thai and Hungarian have the lowest rates of speech and the lowest rates of information conveyed.
- B. Vietnamese conveys information at approximately the same rate as Spanish despite being spoken at a slower rate.



- C. Among the five languages in the table, the language that is spoken the fastest is also the language that conveys information the fastest.
- D. Serbian and Spanish are spoken at approximately the same rate, but Serbian conveys information faster than Spanish does.

Q6

INFORMATION AND IDEAS

The recovery of a 1,000-year-old Chinese shipwreck in the Java Sea near present-day Indonesia has yielded a treasure trove of artifacts, including thousands of small ceramic bowls. Using a portable X-ray fluorescence analyzer tool, Lisa Niziolek and her team were able to detect the chemical composition of these bowls without damaging them. By comparing the chemical signatures of the bowls with those of the materials still at old Chinese kiln sites, Niziolek and her team can pinpoint which Chinese kilns likely produced the ceramic bowls.

Which choice best states the main idea of the text?

- A. Because of a new technology, researchers can locate and recover more shipwrecks than they could in the past.
- B. Researchers have been able to identify the location of a number of Chinese kilns in operation 1,000 years ago.
- C. With the help of a special tool, researchers have determined the likely origin of bowls recovered from a shipwreck.
- D. Before the invention of portable X-ray fluorescence, researchers needed to take a small piece out of an artifact to analyze its components.

Songbirds learn to respond to and imitate their species' songs from an early age. With each generation, small differences are introduced that result in distinct variations—called dialects—among geographically isolated populations of the same species. A research study examined whether twelve-day-old *Ficedula hypoleuca* (pied flycatcher) nestlings prefer local dialects over the unfamiliar dialects of nonlocal *F. hypoleuca* populations: the more begging calls the nestlings made in response to a song, the stronger their preference. The researchers found that nestlings produced more begging calls in response to their own dialect than to nonlocal dialects. Since song preference plays a role in songbird mate selection, the finding suggests that \_\_\_\_\_ blank

Which choice most logically completes the text?

- A. *F. hypoleuca* nestlings' preference for their own dialect likely disappears as they mature to promote socialization between different *F. hypoleuca* populations.
- B. *F. hypoleuca* nestlings who show an early preference for their own dialect are likely to receive more food from their caretakers than nestlings who show no preferences among any *F. hypoleuca* dialects.
- C. *F. hypoleuca* nestlings' preference for their own dialect likely drives them when they mature to reproduce with other *F. hypoleuca* from local rather than nonlocal populations.
- D. *F. hypoleuca* nestlings show a preference for both local *F. hypoleuca* dialects and the songs of other local songbirds over the songs of nonlocal birds of any species.

“To You” is an 1856 poem by Walt Whitman. In the poem, Whitman suggests that he deeply understands the reader, whom he addresses directly, writing, \_\_\_\_\_ blank

Which quotation from “To You” most effectively illustrates the claim?

- A. “Your true soul and body appear before me.”
- B. “Whoever you are, now I place my hand upon you, that you be my poem.”
- C. “I should have made my way straight to you long ago.”
- D. “Whoever you are, I fear you are walking the walks of dreams.”

Choctaw/Cherokee artist Jeffrey Gibson turns punching bags used by boxers into art by decorating them with beadwork and elements of Native dressmaking. These elements include leather fringe and jingles, the metal cones that cover the dresses worn in the jingle dance, a women's dance of the Ojibwe people. Thus, Gibson combines an object commonly associated with masculinity (a punching bag) with art forms traditionally practiced by women in most Native communities (beadwork and dressmaking). In this way, he rejects the division of male and female gender roles.

Which choice best describes Gibson's approach to art, as presented in the text?

- A. He draws from traditional Native art forms to create his original works.
- B. He has been influenced by Native and non-Native artists equally.
- C. He finds inspiration from boxing in designing the dresses he makes.
- D. He rejects expectations about color and pattern when incorporating beadwork.

Dutch painters in the sixteenth and seventeenth centuries often showed tables filled with large wheels of cheese or carved shards of butter. Some art historians, noting that dairy products were a major component of the Dutch diet, interpret these depictions as reflections of everyday Dutch eating habits. However, a group of researchers recently reviewed hundreds of food-related paintings and found that lemons—which could only be acquired in the Netherlands at great cost, since they had to be imported from warmer climates—feature in Dutch paintings of the period more than three times as frequently as dairy products do, thereby casting doubt on the idea that \_\_\_\_\_ blank

Which choice most logically completes the text?

- A. dairy products were a more significant component of the Dutch diet of the period than lemons were.
- B. food was a more popular subject among Dutch painters than it was among painters from other countries at the time.
- C. depictions of food in Dutch paintings of the period should be taken as realistic representations of Dutch eating habits.
- D. Dutch painters of the period may have depicted foods for symbolic reasons rather than to show what Dutch people typically ate.